

Alikhan Mussin

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SUMMARY

I build tools for debugging and incident investigation: a Python SDK that captures and groups application errors, and an NLP-powered platform that classifies incidents and reuses analyst feedback. My projects span backend APIs, SQL persistence, React interfaces, and automated testing with CI.

PROJECTS

MNIST Handwriting Classifier

Personal project · Python, FastAPI, TensorFlow/Keras, CNNs, NumPy, pytest, GitHub Actions, scikit-learn

- Built an end-to-end handwritten digit recognition system with CNN-based classification, custom image preprocessing, and FastAPI inference.
- Improved accuracy on a held-out set of 50 custom handwritten digits from 82% to 90% with data augmentation and to 94% after fine-tuning on a separate 100-image handwriting dataset.
- Exposed predictions through a /predict-image API returning confidence, uncertainty, and Top-3 classes; added automated preprocessing/API tests and GitHub Actions CI.

TraceLite | Error Monitoring SDK & Dashboard

Personal project · Python, FastAPI, SQLAlchemy, SQLite, React, pytest, GitHub Actions

- Captured **100%** of **30** simulated exceptions with stack traces and environment metadata through a Python SDK connected to a FastAPI ingestion API.
- Reduced duplicate error records by **80%** across **100** test events using SHA-256 fingerprints to group repeated exceptions while preserving occurrence counts.
- Made grouped issues and full stack traces inspectable in a React dashboard with application and environment context.
- Automated backend unit and API integration tests with pytest and GitHub Actions on pull requests and pushes to main.

OpsGraph AI | Incident Classification & Semantic Search

Personal project · Python, FastAPI, React, SQLAlchemy, SQLite, Hugging Face Transformers

- Achieved **96,7%** category accuracy and **76,7%** severity accuracy on **60** labeled test incidents by combining zero-shot NLP with domain-specific decision rules.
- Retrieved a relevant historical incident among the top five results for **90%** of **30** test queries using sentence embeddings and semantic similarity search.
- Reused analyst-validated classifications for similar incoming incidents by persisting corrections and prioritizing them in subsequent classification.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

Backend & data: FastAPI, REST APIs, SQLAlchemy, SQLite, MySQL

Frontend: React, HTML, CSS

Testing & delivery: Pytest, GitHub Actions, Git

Applied AI: Hugging Face Transformers, Sentence Transformers, zero-shot classification

EDUCATION

Czech University of Life Sciences

Bachelor of Science | AI/ML Engineer

Prague, Czech Republic

2025 - 2028 May